

## PRACTICAL 3

**AIM :** Automate student marksheet generation, system information display, Fibonacci and prime number generation, and file management operations using shell scripts to enhance computational efficiency and user interaction.

**A) Write a shell script to generate mark- sheet of a student. Take 3 subjects, calculate and display total marks, percentage and Class obtained by the student.**

```
Faiz@PUCKC3F09N9DT MINGW64 /d/OSLab_CM24072/OS_Lab/shell_practical3 (main)
$ nano marksheet.sh

Faiz@PUCKC3F09N9DT MINGW64 /d/OSLab_CM24072/OS_Lab/shell_practical3 (main)
$ chmod +x marksheet.sh
./marksheet.sh
Enter Student Name: Faiz
Enter marks of Subject 1: 65
Enter marks of Subject 2: 77
----- 3: Faiz
Total Marks: h: line 13:
Percentage: 0%
Class: Fail
65: syntax error: operand expected (error token is "
```

**B). Write a menu driven shell script which will print the following menu and execute the given task.**

**Display calendar of current month**

**Display today's date and time**

**Display usernames those are currently logged in the system**

**Display your terminal number**

**1. Display calendar of current month**

```
Faiz@PUCKC3F09N9DT MINGW64 /d/OSLab_CM24072/OS_Lab/shell_practical3 (main)
$ bash system_menu.sh
1. Display current month calendar
2. Display today's date and time
3. Display logged in users
4. Display terminal number
Enter your choice:
1
January 2026
```

## 2. Display today's date and time

```
Faiz@PUCKC3F09N9DT MINGW64 /d/OSLab_CM24072/OS_Lab/shell_practical3 (main)
$ bash system_menu.sh
1. Display current month calendar
2. Display today's date and time
3. Display logged in users
4. Display terminal number
Enter your choice:
2
Tue Jan 20 22:33:21 IST 2026
```

## 3. Display usernames those are currently logged in the system

```
Faiz@PUCKC3F09N9DT MINGW64 /d/OSLab_CM24072/OS_Lab/shell_practical3 (main)
$ bash system_menu.sh
1. Display current month calendar
2. Display today's date and time
3. Display logged in users
4. Display terminal number
Enter your choice:
3
```

## 4. Display your terminal number

```
Faiz@PUCKC3F09N9DT MINGW64 /d/OSLab_CM24072/OS_Lab/shell_practical3 (main)
$ bash system_menu.sh
1. Display current month calendar
2. Display today's date and time
3. Display logged in users
4. Display terminal number
Enter your choice:
4
/dev/pty1
```

C) Write a shell script which will generate first n Fibonacci numbers like: 1, 1, 2, 3, 5, 13

```
Faiz@PUCKC3F09N9DT MINGW64 /d/OSLab_CM24072/OS_Lab/shell_practical3 (main)
$ bash fibonacci.sh
Enter number of terms:
1
Fibonacci Series:
0
```

```
Faiz@PUCKC3F09N9DT MINGW64 /d/OSLab_CM24072/OS_Lab/shell_practical3 (main)
$ bash fibonacci.sh
Enter number of terms:
2
Fibonacci Series:
0 1
```

```
Faiz@PUCKC3F09N9DT MINGW64 /d/OSLab_CM24072/OS_Lab/shell_practical3 (main)
$ bash fibonacci.sh
Enter number of terms:
3
Fibonacci Series:
0 1 1
```

```
Faiz@PUCKC3F09N9DT MINGW64 /d/OSLab_CM24072/OS_Lab/shell_practical3 (main)
$ bash fibonacci.sh
Enter number of terms:
5
Fibonacci Series:
0 1 1 2 3
```

```
Faiz@PUCKC3F09N9DT MINGW64 /d/OSLab_CM24072/OS_Lab/shell_practical3 (main)
$ bash fibonacci.sh
Enter number of terms:
13
Fibonacci Series:
0 1 1 2 3 5 8 13 21 34 55 89 144
```

**D) Write a shell script which will accept a number b and display first n prime numbers as output**

```
Faiz@PUCKC3F09N9DT MINGW64 /d/OSLab_CM24072/OS_Lab/shell_practical3 (main)
$ bash prime.sh
Enter a number:
20
Prime numbers up to 20 are:
2
3
5
7
11
13
17
19
```

**E). Write menu driven program for file handling activity**

**Creation of file**

**Write content in the file**

**Upend file content**

**Delete file content**

```
Faiz@PUCKC3F09N9DT MINGW64 /d/OSLab_CM24072/OS_Lab/shell_practical3 (main)
$ bash file_menu.sh
1. Create File
2. Write to File
3. Append to File
4. Delete File Content
Enter choice:
1
Enter filename:
test1.txt
```

```
Faiz@PUCKC3F09N9DT MINGW64 /d/OSLab_CM24072/OS_Lab/shell_practical3 (main)
$ bash file_menu.sh
1. Create File
2. Write to File
3. Append to File
4. Delete File Content
Enter choice:
2
Enter filename:
test1.txt
Enter content:
Faiz
```

```
Faiz@PUCKC3F09N9DT MINGW64 /d/OSLab_CM24072/OS_Lab/shell_practical3 (main)
$ bash file_menu.sh
1. Create File
2. Write to File
3. Append to File
4. Delete File Content
Enter choice:
3
Enter filename:
test1.txt
Enter content:
Faiz
```

```
Faiz@PUCKC3F09N9DT MINGW64 /d/OSLab_CM24072/OS_Lab/shell_practical3 (main)
$ bash file_menu.sh
1. Create File
2. Write to File
3. Append to File
4. Delete File Content
Enter choice:
4
Enter filename:
test1.txt

Faiz@PUCKC3F09N9DT MINGW64 /d/OSLab_CM24072/OS_Lab/shell_practical3 (main)
$ cat test1.txt
```